

Can Quantum-Mechanical Description of Physical Reality be Considered Complete?

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1. ABSTRACT & THEORETICAL FOUNDATION

In a complete physical theory there is an element corresponding to each element of reality. We demonstrate that wave mechanics predicts non-local correlations between spatially separated quantum systems.

2. CORE CONCEPTUAL INTUITION (PLAIN ENGLISH)

"Two entangled coins always land on the exact same face even if you flip one on Earth and the other on Mars, proving nature is non-locally connected."

3. Mathematical Formulation & Unitary Rule

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle), \quad \text{CHSH Bound: } 2\sqrt{2}$$

Measuring one qubit instantly projects the second qubit into the matching outcome with 100% correlation regardless of spatial separation.

4. Step-by-Step Circuit Sequence

Produced using a Hadamard gate on qubit 0 followed by a CNOT gate targeting qubit 1.

5. Executable IBM Qiskit (Python) Code

```
from qiskit import QuantumCircuit

qc = QuantumCircuit(2, 2)
qc.h(0)      # Superposition on control qubit
qc.cx(0, 1)  # Entangle control with target
qc.measure([0, 1], [0, 1])

print(qc.draw(output='text'))
```

6. Computational Complexity & Speedup

Classical Time:

Quantum Time:

Speedup Class:

Requires 2 classical communication rounds. Instantaneous state collapse (non-local signaling without classical communication).

7. Key Applications & Future Research Directions

- Quantum Key Distribution (E91 protocol)
- Quantum Teleportation & Superdense Coding
- Distributed quantum computing networks

Future Work:

Experimental testability via Bell inequalities, device-independent cryptography, and multi-partite entanglement.